Interval Estimation: Uniform Distribution

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Let's suppose that we sample $n = 10$ independent and identically distributed (iid) data from a $\text{Uniform}(0, \theta)$ distribution:

```
set.seed(101)
n     <- 10
theta <- 1.2
X     <- runif(n,min=0,max=theta)

sort(X)
```

```
##  [1] 0.05258978 0.29982687 0.36006580 0.40016057 0.4466380
5 0.65499427
##  [7] 0.70183995 0.74641436 0.78922848 0.85162082
```

What is a 95% one-sided lower bound on θ? (More to the point: how would students determine this bound?)

1. Specify the statistic that will be used to derive the bound.

When possible, we should utilize a *sufficient statistic*…does one exist here?

The pdf for the $\text{Uniform}(0, \theta)$ distribution is

$$f_X(x) = \frac{1}{\theta - 0} = \frac{1}{\theta} \qquad x \in [0, \theta] \,,$$

and thus the likelihood of θ, given the observed data $\mathbf{x}$, is

$$L(\theta|\mathbf{x}) = \prod_{i=1}^{n} \frac{1}{\theta} = \frac{1}{\theta^n} .$$

We note that no function of the data appears in the likelihood expression. So as we have to do with distributions that have domain-specifying parameters, we explicitly include an indicator function:

$$f_X(x) = \frac{1}{\theta} \mathbb{1}_{(x \in [0,\theta])} \quad \Rightarrow \quad L(\theta|\mathbf{x}) = \frac{1}{\theta^n} \prod_{i=1}^{n} \mathbb{1}_{(x_i \in [0,\theta])} .$$

The likelihood function is zero unless θ is larger than (or equal to) the largest datum. Thus a sufficient statistic for θ is $X_{(n)}$, the maximum observed datum.

2. What is the cumulative distribution function for the sampling distribution for the statistic?

The maximum datum has the probability density function

$$f_{(n)}(x) = n f_X(x) [F_X(x)]^{n-1}$$

and the cumulative distribution function

$$F_{(n)}(x) = [F_X(x)]^n .$$

Thus we need to derive the cdf for the individual data:

$$F_X(x) = \int_0^x \frac{y}{\theta} dy = \left. \frac{y}{\theta} \right|_0^x = \frac{x}{\theta} ;$$

given this, we can see that

$$F_{(n)}(x) = \left(\frac{x}{\theta} \right)^n .$$

3. What is the interval bound?

To determine the interval bound, we solve the equation

$$F_Y(y_{\rm obs} \vert \theta) - q = 0$$

for θ, where Y is our chosen statistic, $F_Y(\cdot)$ is the cdf for the sampling distribution for Y, $y_{\rm obs}$ is the observed statistic value, and q is an appropriately chosen quantile (see below).

Here, the equation becomes

$$\left(\frac{X_{(n)}}{\theta}\right)^n - q = 0 \,,$$

or

$$\hat{\theta} = \frac{X_{(n)}}{q^{1/n}} \,.$$

But…what is q? The confidence interval reference table is

Interval Type	Does $E[Y]$ Increase with θ?	q for Lower Bound	q for Upper Bound
two-sided	yes	$1 - \alpha/2$	$\alpha/2$
two-sided	no	$\alpha/2$	$1 - \alpha/2$
one-sided lower	yes	$1 - \alpha$	-
one-sided lower	no	α	-
one-sided lower	yes	-	α
one-sided lower	no	-	$1 - \alpha$

where the confidence coefficient is $1 - \alpha$ (so that α here is 0.05). We need not explicitly compute $E[X_{(n)}]$ to know that its value increases as θ increases. However, if we needed to:

$$E[X_{(n)}] = \int_0^\theta x f_{(n)}(x) dx = \int_0^\theta x F'_{(n)}(x) dx = \int_0^\theta x \frac{n}{\theta} \left(\frac{x}{\theta} \right)^{n-1} dx$$
$$= \frac{n}{\theta^n} \int_0^\theta x^n dx = \frac{n}{n+1} \frac{1}{\theta^n} x^{n+1} |_0^\theta = \frac{n}{n+1} \theta \,.$$

So we are on the third line of the table: $q = 1 - \alpha = 0.95$.

So now, putting it all together in `R` code:

```
n <- 10
q <- 0.95
X.n <- max(X)
cat("The maximum observed value is   ",round(X.n,3),"\n")
```

```
## The maximum observed value is    0.852
```

```
cat("The one-sided lower 95% bound is",round(X.n/(q^(1/n)),
3),"\n")
```

```
## The one-sided lower 95% bound is 0.856
```